



SEAEDD Southeast Arkansas Economic
Development District



FY 2023–FY 2024: CONSOLIDATED RAIL INFRASTRUCTURE
AND SAFETY IMPROVEMENTS PROGRAM (CRISI)

May 28, 2024

DELTALINK REGIONAL RAIL

Multimodal Freight Connectivity for the Mississippi Delta Region

Benefit-Cost Analysis Technical Memorandum



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Executive Summary

The Southeast Arkansas Economic Development District (SEAEDD), Lake Providence Port Commission (LPPC), the Chicot-Desha Metropolitan Port Authority (Yellow Bend Port), and the North Louisiana & Arkansas Railroad (NLA) are continuing their partnership to re-develop a multimodal freight corridor located along a 150-mile-long section of the Mississippi River (the designated M-55 Marine Highway) between McGehee, Arkansas, south to Vidalia, Louisiana. The DeltaLink Regional Rail Project (the Project) is a unique public-private partnership (P3) between a state economic development agency (SEAEDD), four public ports (Port of Lake Providence, Tensas Parish Port, Vidalia Port, and Yellow Bend Port), and a private short line railroad (NLA). Located in one of the most severely economically distressed rural regions of the US, the Project is located in Desha and Chicot Counties, AR, and in East Carroll, Madison, Tensas, and Concordia Parishes, LA.

This section of the Mississippi River between McGehee and Vidalia is also served by a parallel continuous short line railroad corridor that interfaces directly with the existing Port of Lake Providence and Madison Parish Port and yet-to-be constructed rail connections to the separate ports of Yellow Bend near Arkansas City, AR, Tensas Parish Port near St. Joseph, LA, and Vidalia Port; several local railroad shippers; and three other railroads with local, regional, national, and international connections. US Highway 65 (US 65) parallels the McGehee-Tallulah Railroad Corridor.

Much of the McGehee-Vidalia Railroad Corridor has been closed or underutilized for approximately 15 years. During this period there was a decrease in demand for freight rail service, which is consistent with the overall economic decline of the region. Coupled with overall neglect, this left the rail line in poor condition. Eventually, rail service ceased due to difficulties operating over substandard rail infrastructure. In an effort to save the corridor from abandonment, the Southeast Arkansas Economic Development District and Lake Providence Port Commission collectively acquired 60 miles of the track between 2008 and 2011. Their goal was to preserve rail service, spur economic development, and provide transportation connectivity with the Class I railroads. The North Louisiana and Arkansas Railroad now operates regular rail service along this portion of the corridor.

This Consortium of project partners are currently working together on the SEAEDD Southeast Arkansas and Northeast Louisiana Multimodal Freight Corridor Improvement Project, partially funded by an INFRA Grant. The rail line between McGehee and Lake Providence Port is back in service and the track rehabilitation work will continue throughout 2024. **Figure 1** below shows the previous project's track location in purple and current project's track locations are in multiple colors shown in the legend.

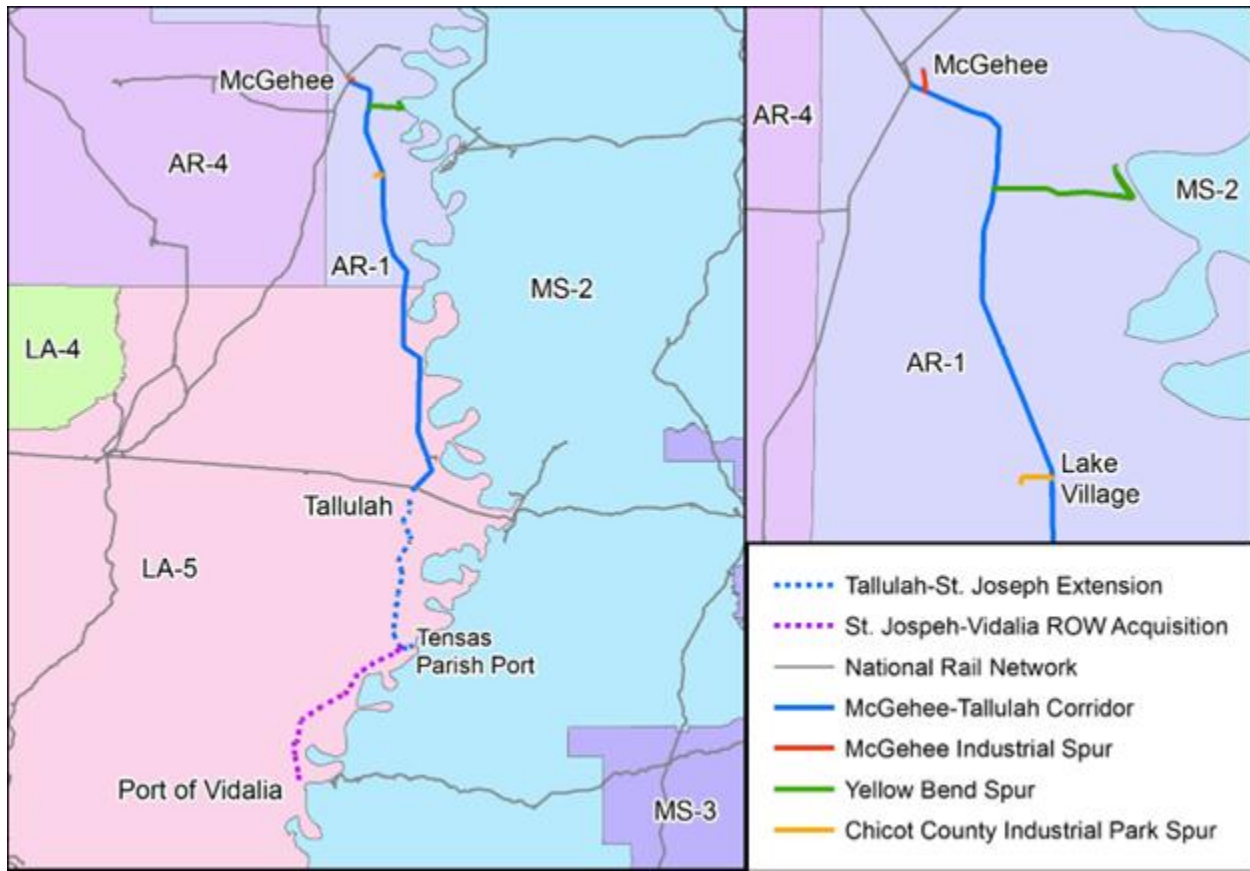


Figure 1: SEAEDD DeltaLink Project – Project Location

A Benefit-Cost Analysis (BCA) of the Project was conducted in conformance with federal guidance regarding evaluation methods and monetization values recommended by the U.S. Department of Transportation (U.S. DOT) in its December 2023 *Benefit-Cost Analysis Guidance for Discretionary Grant Programs*. A table summarizing the changes expected from the Project (and the associated benefits) is provided below.

Table ES - 1: Summary of Infrastructure Improvements and Associated Benefits

Current Status (Base Scenario) & Problems to be Addressed	Changes to Baseline (Alternative Scenario)	Types of Impact	Population Affected by Impacts	Economic Benefits	Summary of Results (2022\$, Discounted)
Currently, the Ports of Yellow Bend, Tensas, and Vidalia do not connect to a rail line. There is also no track from Tallulah, LA to Vidalia, LA.	The Project would add a connection from the Port of Yellow Bend to existing rail and add new rail to connect the Port of Tensas to existing rail. ROW acquisition, NEPA, and preliminary engineering would be conducted to connect the Port of Vidalia. The north end of the track would connect to existing Union Pacific rail line and the south end to existing Kansas City Southern rail line.	Improved safety and avoided accident costs from reduced freight truck movements.	Local motorists	Avoided Freight Transportation Accidents	\$121 M
		Avoided emissions costs from avoided truck movements.	Residents within the region	Avoided Emissions	\$102 M
		Avoided fuel costs	Shippers	Avoided Fuel Costs	\$18 M
		Avoided congestion costs	Local and State residents and motorists	Avoided Roadway Congestion Costs	\$12 M
		Shipper cost efficiencies from truck to rail diversion	Shippers	Reduced Shipping Costs	\$1,466 M
		Avoided pavement maintenance cost from avoided truck movements.	Local and State governments	Avoided Pavement Damage Costs	\$14 M
		Additional O&M from construction of new track	Project sponsors	Increased Maintenance Costs	-\$13 M
		Reduced noise pollution from less truck traffic	Local and State residents	Reduced noise pollution	n/a
		Increased Opportunities for Development	Local and State residents, Shippers, residents within the region	Growth opportunities for local businesses	n/a

The period of analysis used in the estimation of benefits and costs corresponds to 35 years, including 5 years of design, engineering, and construction as well as 30 full years of operation. The capital cost is estimated at \$172 million in capital expenditures, in 2022 dollars. The capital expenditure for this Project by year is presented in **Table ES - 2**, while the total benefits, by merit criteria, are presented in **Table ES - 3**.

Table ES - 2: Summary of Capital Expenditure, 2022 Dollars

Year	Capital Expenditure
2025	\$2,425,000
2026	\$4,462,000
2027	\$32,610,333
2028	\$105,997,332
2029	\$26,499,333

Table ES - 3: Summary of Benefits by Merit Criteria, 2022 Dollars

Merit Criteria	Benefit Category	Description	Monetized	Quantitative
Safety	Avoided Freight Transportation Accidents	Eliminating potential accidents generated by transporting commodities by truck.	Yes	\$121 M
Environmental Sustainability	Avoided GHG Emissions	Avoided emissions costs from avoided truck movements.	Yes	\$29 M
	Avoided CAC Emissions			
Economic Competitiveness	Avoided Fuel Costs	Avoided fuel costs	Yes	\$18 M
	Avoided Roadway Congestion Costs	Avoided congestion costs	Yes	\$12 M
	Avoided Additional Freight Transportation Costs	Avoided freight transportation costs from transporting goods using truck instead of rail.	Yes	\$1,466 M
State of Good Repair	Avoided Pavement Damage Costs	Avoided pavement maintenance cost from avoided truck movements.	Yes	\$14 M
	Avoided Maintenance Costs	Avoided additional O&M due to reconstruction of bridge	Yes	-\$13

The relevant data and calculations used to derive the benefits and costs of the Project are provided in the BCA Model submitted with this application. Based on the BCA conducted (see **Table ES - 4**), the Project is expected to generate \$1,720 million in discounted benefits, while costing \$143.8 million (discounted) based on a 2.0 percent real discount rate for carbon dioxide (CO₂)-related impacts and a 3.1 percent real discount rate for all other impacts (as per U.S. DOT BCA guidance). This translates to a net present value (NPV) of \$1,576 million (discounted) and a benefit-cost ratio of 12.0. Additional detailed breakdowns of the analysis, including the various assumptions and methodologies, are presented in the rest of this document.

Table ES - 4: Overall Results of the Benefit-Cost Analysis, 2022 Dollars

Evaluation Metrics	Undiscounted	Discounted
Total Benefits	\$3,400 M	\$1,720 M
Total Costs	\$172 M	\$144 M
Net Present Value (NPV)	\$3,228 M	\$1,576 M
Return on Investment (ROI)	1877%	1096%

Evaluation Metrics	Undiscounted	Discounted
Benefit-Cost Ratio (BCR)	19.8	12.0
Payback Period (years)	9.0	10.0
Internal Rate of Return (IRR)	28.9%	

In addition to the monetized benefits summarized above, the Project would generate transportation benefits that are difficult to quantify. A brief description of this is provided below.

- **Reduction in Noise Pollution**

As railcars can load larger volumes than freight trucks, the mode shift from freight truck to rail reduces the volume of freight trucks and their associated noise pollution.

- **Additional Development Opportunities**

The availability of an additional mode of transportation increases the opportunities for economic development. Providing industrial-type businesses with expanded opportunities to transport goods using rail, a typically lower cost mode than truck, increases the viability of expansion projects or new developments.

Introduction

This document provides detailed technical information on the economic analyses conducted in support of the U.S. Department of Transportation (U.S. DOT) Fiscal Year (FY) 2023 - 2024 Consolidate Rail Infrastructure and Safety Improvements (CRISI) Program

for the Southeast Arkansas Economic Development District (SEAEDD) DeltaLink Regional Rail Project (Project). This document includes the following.

- **Methodological Framework:** Introduces the conceptual framework used in the Benefit-Cost Analysis (BCA).
- **Overview:** Provides an overview of the Project, including a brief description of existing conditions and proposed alternatives; a summary of cost estimates and schedule; and a description of the types of effects that the Project is expected to generate.
- **General Assumptions:** Discusses the general assumptions used in the estimation of project costs and benefits.
- **Demand Projections:** Provides estimates of travel demand and traffic volumes.
- **Benefits Measurement, Data, and Assumptions:** Details the specific data elements and assumptions used to address the goals of the Project to comply with MPDG program requirements.
- **Summary of Findings and Benefit-Cost Outcomes:** Provides estimates of the net present value (NPV), its benefit-cost ratio (BCR), and other evaluation metrics.
- **Benefit Cost Analysis Sensitivity:** Provides the outcome of the sensitivity analysis that evaluates the difference assumptions made in the analysis, and the impact that the variability of those assumptions may have on the overall results.

Methodological Framework

The BCA conducted for this Project includes monetized benefits and costs measured using U.S. Department of Transportation (U.S. DOT) guidance, *Benefit-Cost Analysis Guidance for Discretionary Grant Programs* (BCA Guidance) updated in December 2023, as well as the quantitative and qualitative merits of the Project. A BCA provides estimates of the benefits that are expected to accrue over a specified period and compares them to the anticipated costs. Costs include both the resources required to develop the Project and the costs of maintaining the new or improved asset over time. Estimated benefits are based on the projected impacts of the Project on both users and non-users of the facility, valued in monetary terms.¹

While a BCA is just one of many tools that can be used in making decisions about infrastructure investments, U.S. DOT believes that it provides a useful benchmark from which to evaluate and compare potential transportation investments.²

¹ U.S. DOT, *Benefit-Cost Analysis Guidance for Discretionary Grant Programs*, December 2023

² *Ibid.*

The specific methodology employed for this application was developed using the BCA guidance developed by U.S. DOT and is consistent with the MPDG program guidelines. In particular, the methodology involves:

- Establishing existing and future conditions under the Base Case (No Build) and Alternative Case (Build) scenarios;
- Assessing benefits with respect to each of the merit criteria identified in the Notice of Funding Opportunity (NOFO);
- Measuring benefits in dollar terms, whenever possible, and expressing benefits and costs in a common unit of measurement;
- Using U.S. DOT guidance for the valuation of safety benefits and reductions in air emissions, while relying on industry best practice for the valuation of other effects;
- Discounting future benefits and costs with the real discount rates recommended by the U.S. DOT (3.1 percent except for CO₂-related impacts, which are discounted at 2.0 percent); and
- Conducting a sensitivity analysis to assess the impacts of changes in key assumptions.

Overview

The Project's location is spread across multiple counties in Southeast Arkansas and Northeast Louisiana.

By adding rail connections for two existing ports to track currently being rehabilitated, the Project will:

- Lower shipping and roadway maintenance costs by adding the use of rail transportation.
- Reduce road congestion, noise from trucks, and emissions by opening up alternative modes for goods traveling to and from the ports.
- Maintain road safety by adding rail transportation as a modal option for freight shippers.
- Provide existing ports with growth opportunities utilizing rail as the shipping mode.

- Increase development opportunities for other areas along the newly built track in Northeast Louisiana.

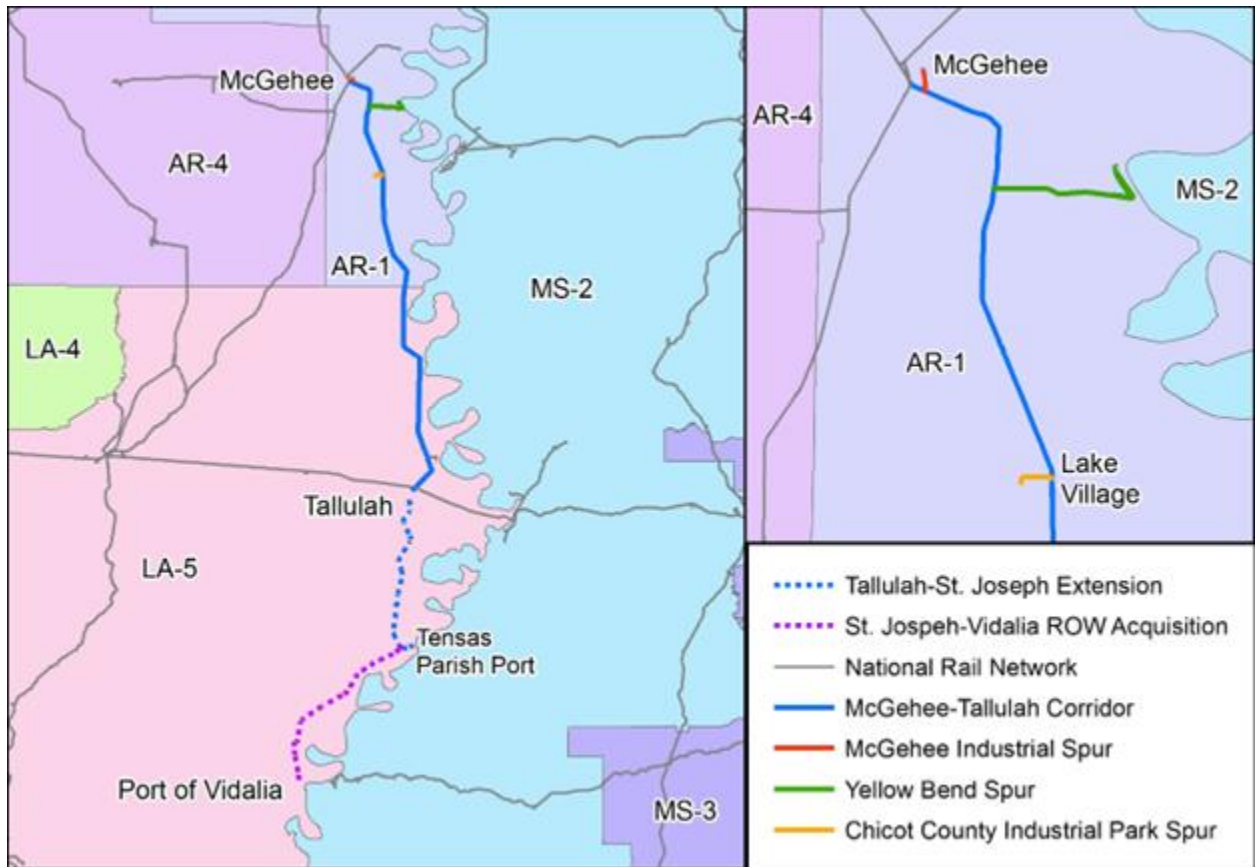


Figure 2 highlights the Project's location.

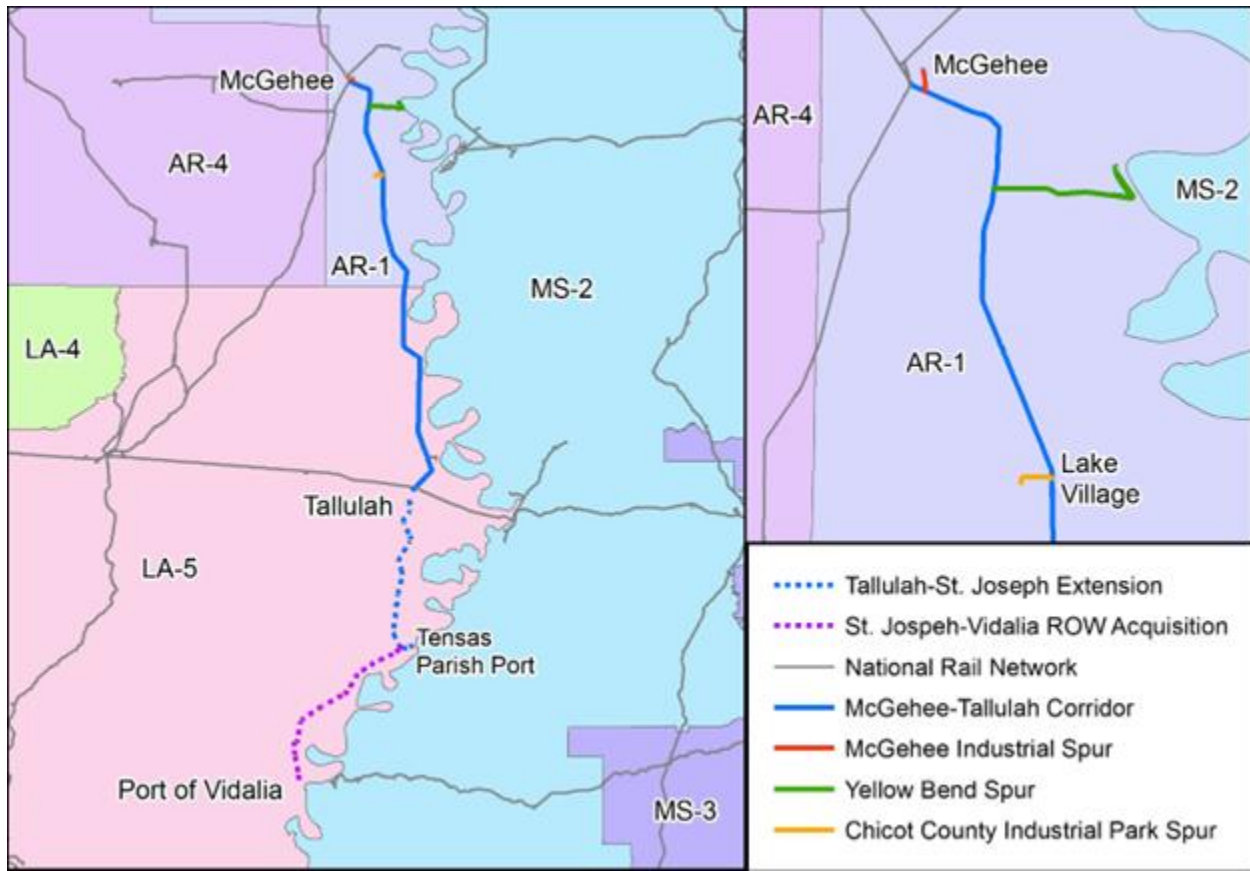


Figure 2: SEAEDD DeltaLink Project – Project Location

Base Case

The Base Case for the Project is defined as the No Build scenario. In the Base Case, other modes of transportation are used to move goods to and from the ports. There are also other development opportunities in the area which might not come to fruition if the rail line is not available.

Alternative Case

The Alternative Case is defined as the Build scenario. In the Alternative Case, rail connections are added to two ports and track is extended farther south from an existing line. Thus, under the Build scenario, there are additional modes available to transport goods. This would translate to avoided emissions, pavement damage, freight costs, fuel costs, and potential accidents from additional roadway traffic. NEPA, ROW acquisition, and preliminary engineering will be performed on a route that would connect a third port. As the track will not be built under this project, benefits from connecting the third port are not included in this analysis.

Project Cost and Schedule

The capital costs would be incurred starting in 2025 through completion in 2029. The line will be open to rail traffic in 2029. However, to be conservative benefits will not start accruing until 2030. The annual expenditures are summarized in **Table 1**, while the cost by component are presented

in **Table 2**. Costs are shown in 2023 and 2022 dollars to show connection with 2023 values presented in narrative. Costs were deflated by 0.97 based on the Bureau of Economic Analysis Implicit Price Deflators for Gross Domestic Product.

Table 1: Expenditure Profile, 2022 Dollars

Year	Capital Expenditure
2025	\$2,425,000
2026	\$4,462,000
2027	\$32,610,333
2028	\$105,997,332
2029	\$26,499,333

Table 2: Capital Cost by Component, 2023 and 2022 Dollars

Component	Capital Cost, 2023 \$	Capital Cost, 2022 \$	Percentage of Cost
NEPA and PE	\$5,000,000	\$4,850,000	2.82%
Final Design	\$2,400,000	\$2,328,000	1.35%
Property Acquisition	\$6,000,000	\$5,820,000	3.38%
Construction Mgmt	\$3,600,000	\$3,492,000	2.03%
Construction	\$160,313,400	\$155,503,998	90.41%
Total	\$177,313,400	\$171,993,998	100.00%

Merit Criteria Impacts

The benefits associated with the Project are presented in **Table 3**, and they are separated by the various merit criteria.

Table 3: Expected Effects on Benefit Categories, Discounted

Merit Criteria	Benefit Category	Description	Monetized	Quantitative
Safety	Avoided Freight Transportation Accidents	Eliminating potential accidents generated by transporting commodities by truck.	Yes	\$121 M
Environmental Sustainability	Avoided GHG Emissions	Avoided emissions costs from avoided truck movements.	Yes	\$29 M
	Avoided CAC Emissions			
Economic Competitiveness	Avoided Fuel Costs	Avoided fuel costs	Yes	\$18 M
	Avoided Roadway Congestion Costs	Avoided congestion costs	Yes	\$14 M
	Avoided Additional Freight Transportation Costs	Avoided freight transportation costs from transporting goods using truck instead of rail.	Yes	\$1,466 M
State of Good Repair	Avoided Pavement Damage Costs	Avoided pavement maintenance cost from avoided truck movements.	Yes	\$14 M
	Avoided Maintenance Costs	Avoided additional O&M due to reconstruction of bridge	Yes	-\$13

General Assumptions

The BCA measures benefits against costs throughout a period of analysis, beginning at the start of construction and including 30 full years of operations.

The monetized benefits and costs are estimated in 2022 dollars, with future dollars discounted in compliance with U.S. DOT CRISI requirements.

The methodology makes several important assumptions and seeks to avoid overestimation of benefits and underestimation of costs. Specifically:

- Input prices are expressed in 2022 dollars;
- The period of analysis begins in 2025 and ends in 2059; it includes project development and construction years (2025–2029) and 30 full years of operations (2030–2059); and
- A constant 2.0 percent real discount rate for carbon dioxide (CO₂)-related benefits and 3.1 percent real discount rate for all other benefits are assumed throughout the period of analysis.

Demand Projections

Accurate demand projections are important to ensure reasonable BCA output results. The magnitudes of the long-term benefits accruing over the study period are a function of the volume of freight transported by rail in the project area, and the diversion distances by mode. The methodology used to calculate incremental freight volumes and diversion distances is detailed below.

Methodology

The demand for freight movements and the associated truck (Base Case) and train (Alternative Case) volumes were provided by the Ports benefiting from improved access to rail. Some of the specific data collected for shippers included:

- Estimate of truckloads per year required to ship all products;
- Estimate of rail carload-equivalents required to ship all freight by train;
- Origin and destination pairs of goods being shipped and the associated distance of haul;
- Shipping costs by truck and carload.

While some shippers suggested they would haul goods by truck over long distances, it was determined that beyond a distance of approximately 500 miles, it becomes more economical to truck goods to a transload facility and ship the remainder of the distance by rail. Given that the average distance to the nearest transload facilities for shippers in the project vicinity was assessed at 27 miles, it was assumed that only origins/destinations within 500 miles would be trucked directly to market. For any movements over longer distances would instead be trucked 27 miles and moved by rail from there. Given that the remainder of the haul would be by rail and equivalent in the Base and Alternative Cases, only the trucked component is considered for analysis.

A detailed analysis of the U.S. Federal Highway Administration's (FHWA) Freight Analysis Framework (FAF4) forecasts for specific commodities and origin-destination pairs was used to generate a long-term freight movement growth rate.

The diversion of trucks to rail is a key driver of public value for the Project, so freight demand forecasts and modal diversion estimates were derived from shipper consultations and market analysis. Several assumptions were made as part of the analysis, with some of the key ones stipulating that:

- All volumes considered for analysis cannot currently be moved by rail and would have to be moved by truck as the next-best alternative.
- New infrastructure proposed under the Alternative Case would allow all identified shippers to move all freight by rail.

Assumptions

All assumptions used in the estimation of demand inputs for the Project are provided in.

Table 4: Assumptions, Ports of Yellow Bend and Tensas

Input Assumption	Source
Average Freight per Truck is 25 tons/truck	Assumption in line with maximum Interstate Highway load limits
Average Freight per Train is 10k tons/train	Based on shipper data for types of commodities and industry averages.
Average cost of shipping by Truck is \$0.208/ton-mile in 2022 \$	BLS Table 3-21: Average Freight Revenue Per Ton-mile (Current cents), accessed 2022 dollars.
Average cost of shipping by Rail is \$0.019/ton-mile in 2022 \$	BLS Table 3-21: Average Freight Revenue Per Ton-mile (Current cents), accessed 2022 dollars.
Train fuel efficiency is 500 ton-miles/gallon	Average reported by the Association of American Railroads
Truck fuel efficiency is 11.5 miles/gallon	Maximum found in Annual Fleet Fuel Study conducted by the North American Council for Freight Efficiency
Pavement Maintenance cost for rural interstate trucks in 2022\$ is \$0.09/truck mile	Addendum to the 1997 Federal Highway Cost Allocation Study, Final Report, U.S. Department of Transportation and Federal Highway Administration, May 2000; Table 13. Assuming 60 kip 5-axle trucks 65% on rural and 35% on urban highways, escalated to 2022\$.

Demand Projections

The resulting projections for freight volumes are presented in **Table 5** and **Table 6** for Port of Yellow Bend and **Table 7** and **Table 8** for Port of Tensas. Further details are available in the accompanying BCA spreadsheet model. Projections are based on shipper surveys and existing production facilities' proximity to each port.

Table 5: Port of Yellow Bend Demand Projections

Commodity	Tons/year	Distance from Port
Wood Chips	18,000	75
New wood/timber	128,700	68
Grain	487,500	188
Soybeans & rice	8,750	200
Total	642,950	530

Table 6: Port of Yellow Bend Assumed Quantities and Distance of Freight Diversion

Commodity	Tons/year	Distance from Port
Wood Chips	9,000	38
New wood/timber	64,350	34
Grain	243,750	94
Soybeans & rice	4,375	100
Total	321,475	265

Table 7: Port of Tensas Demand Projections

Commodity	Tons/year	Distance from Port
Aggregate	107,000	40
Corn	250,000	65
Soybeans	100,000	65
Cottonseed	20,000	65
Equipment	1,000	860
Fertilizer	37,000	40
Total	515,000	1,135

Table 8: Port of Tensas Assumed Quantities and Distance of Freight Diversion

Commodity	Tons/year	Distance from Port
Aggregate	53,500	20
Corn	125,000	33
Soybeans	50,000	33
Cottonseed	10,000	33
Equipment	500	430
Fertilizer	18,500	20
Total	257,500	568

Benefits Measurement, Data, and Assumptions

This section describes the measurement approach used for each benefit or impact category identified in **Table 3** and provides an overview of the associated methodology, assumptions, and estimates.

Safety

Accident costs and impacts on life, limb, and property are a significant component of user costs for transportation. Transportation safety is a key economic factor in the planning of the movement of goods and people, as well as an important indicator of transportation efficiency. Beyond the economic context, transportation safety is often the subject of public concern.

The Project would help achieve U.S. DOT's long-term safety outcome through avoiding incremental miles driven by trucks and rail in the No Build scenario. Freight truck traffic has a higher rate of injury and fatality accidents compared to freight rail as shown in **Table 10**. Values

for fatalities and injuries from freight rail and truck traffic were obtained from a Texas A&M Transportation Institute study.³

Avoided Freight Transportation Accidents

METHODOLOGY

Since the Project will avoid transporting freight between Fairbanks and other locations by truck, it will also avoid potential additional accidents during such transport. This stems from the fact that transporting freight by train is generally safer than transporting freight by truck. The accidents from freight transportation that would be avoided are calculated based on estimates of annual ton-miles transported by mode in the No Build and Build scenarios and the accident rates by severity and mode. Estimates of expected accidents by severity are then monetized using the respective cost per accident severity provided by U.S. DOT's BCA guidance. Therefore, the avoided freight transportation accidents benefit is the difference between the expected freight accident costs in the No Build and Build scenarios.

ASSUMPTIONS

The assumptions used in the estimation of freight transportation accidents are presented in **Table 9** and **Table 10**.

Table 9: Assumptions used to Monetize Safety Benefits

Variable	Units	Value	Source
Cost of Fatality	2022\$/fatality	\$12,500,000	US Department of Transportation, Benefit-Cost Analysis Guidance for Discretionary Grant Programs, December 2023.
Cost of Injury (Unknown Severity)	2022\$/injury	\$217,600	

Table 10: Assumptions Used in the Estimation of Freight Transportation Accidents

Variable	Units	Value	Source
Fatalities - Truck	fatalities/billion ton-mile	2.221	Texas A&M Transportation Institute. A Modal Comparison of Domestic Freight Transportation Effects on the General Public: 2001–2009. Published in 2013 ⁴
Injuries - Truck	injuries/billion ton-mile	55.171	
Fatalities - Freight Rail	fatalities/billion ton-mile	0.479	
Injuries - Freight Rail	injuries/billion ton-mile	4.621	

Estimate of Safety Benefits

Table 11 provides the monetized benefit estimates of safety benefits. The estimated present value of the discounted benefits over the projected 30-year benefit period is \$121 million.

Table 11: Estimates of Safety Benefits

Values in 2022\$	Undiscounted	Discounted
Avoided Freight Transportation Accidents	\$242 M	\$121 M
Total	\$242 M	\$121 M

³ Texas A&M Transportation Institute. A Modal Comparison of Domestic Freight Transportation Effects on the General Public: 2001–2019. January 2022.

⁴ <https://scholars.library.tamu.edu/vivo/display/n320381SE/Documents/View%20All>

Environmental Benefits

Environmental costs are increasingly considered an important component in the evaluation of transportation projects. The primary environmental impact of vehicle use is exhaust emissions, which impose wide-ranging social costs on people, materials, and vegetation. The negative effects of pollution depend not only on the quantity of pollution produced, but also on the types of pollutants emitted and the local environmental conditions into which the pollution is released.

The Project is expected to provide environmental benefits by mitigating ton-miles driven by trucks due to diversions while the bridge is out of service. Emissions are reduced in the Build scenario relative to the No Build Scenario due to transporting freight through a more efficient mode.

The assumptions used to monetize the reduction in emissions are summarized in **Table 12**.

Table 12: Assumptions Used in the Estimation of Environmental Benefits – Emission Values

Emission Type	NOx	SO ₂	PM _{2.5} **	CO ₂	Source/Comment
2021	\$19,800	\$52,900	\$951,000	\$228	Technical Support Document: Estimating the Benefit per Ton of Reducing PM _{2.5} Precursors from 17 Sectors (February 2018)" https://www.epa.gov/sites/default/files/2018-02/documents/sourceapportionmentbptsd_2018.pdf
2022	\$19,800	\$52,900	\$951,000	\$228	
2023	\$19,800	\$52,900	\$951,000	\$228	
2024	\$20,100	\$53,800	\$963,200	\$233	
2025	\$20,300	\$54,800	\$975,500	\$237	
2026	\$20,600	\$56,100	\$993,500	\$241	NO _x , SO _x , and PM _{2.5} values are inflated from 2015 to 2022 dollars using the GDP deflator. CO ₂ values are inflated from 2020 to 2022 dollars using the GDP deflator. EPA Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances (November 2023)
2027	\$21,000	\$57,400	\$1,011,900	\$245	
2028	\$21,300	\$58,700	\$1,030,600	\$250	
2029	\$21,700	\$60,100	\$1,049,600	\$253	
2030	\$22,000	\$61,500	\$1,069,000	\$257	
2031	\$22,000	\$61,500	\$1,069,000	\$262	
2032	\$22,000	\$61,500	\$1,069,000	\$265	
2033	\$22,000	\$61,500	\$1,069,000	\$270	
2034	\$22,000	\$61,500	\$1,069,000	\$274	
2035	\$22,000	\$61,500	\$1,069,000	\$278	
2036	\$22,000	\$61,500	\$1,069,000	\$282	Note: Fuel saved (gasoline, diesel, natural gas, etc.) can be converted into metric tons of emissions using EPA guidelines available at https://www.epa.gov/energy/greenhouse-gases-equivalencies-calculatorcalculations-and-references
2037	\$22,000	\$61,500	\$1,069,000	\$287	
2038	\$22,000	\$61,500	\$1,069,000	\$290	
2039	\$22,000	\$61,500	\$1,069,000	\$294	
2040	\$22,000	\$61,500	\$1,069,000	\$299	
2041	\$22,000	\$61,500	\$1,069,000	\$303	
2042	\$22,000	\$61,500	\$1,069,000	\$308	
2043	\$22,000	\$61,500	\$1,069,000	\$312	
2044	\$22,000	\$61,500	\$1,069,000	\$317	
2045	\$22,000	\$61,500	\$1,069,000	\$321	
2046	\$22,000	\$61,500	\$1,069,000	\$326	Note: The recommended values for reducing CO ₂ emissions reported in Table A-6 represent the values of future economic damages that can be avoided by reducing emissions in each future year by one metric ton. After using per metric ton values to estimate the total value of reducing CO ₂ emissions in any future year, the result must be further discounted to its present value as of the analysis year used in the BCA, also using a 2.0 percent discount rate
2047	\$22,000	\$61,500	\$1,069,000	\$331	
2048	\$22,000	\$61,500	\$1,069,000	\$336	
2049	\$22,000	\$61,500	\$1,069,000	\$340	
2050	\$22,000	\$61,500	\$1,069,000	\$345	

Avoided Freight Emissions

METHODOLOGY

The avoided emissions from freight transportation considers the change in emissions from transporting freight by truck versus by rail. Specifically, this benefit is derived by the emission factors by mode for each of CO₂, NO_x, VOC, PM_{2.5}, and SO₂ for both the No Build and Build

scenarios. To obtain the expected emission costs, the volume of each pollutant was then converted to tons and multiplied by its monetary value (per metric ton), as presented in **Table 12**.

ASSUMPTIONS

While the emission factors used in the estimation of environmental benefits are summarized in **Section 0** and **Section 0** respectively, the assumptions used to estimate the gallons of fuel consumed for freight rail transportation are presented in **Table 13**.

Table 13: Assumptions Used in the Estimation of Environmental Benefits – Fuel Consumption Rate

Variable	Units	Value	Source
Train Fuel Efficiency	ton-miles/gallon	500	Average reported by the Association of American Railroads ⁵
Truck Fuel Efficiency	miles/gallon	11.5	Maximum found in Annual Fleet Fuel Study conducted by the North American Council for Freight Efficiency ⁶

Estimate of Environmental Benefit

Table 14 shows the estimated environmental benefits. The estimated discounted present value over the projected 30-year benefit period is about \$102 million.

Table 14: Estimates of Environmental Benefits

Values in 2022\$	Undiscounted	Discounted
Avoided GHG Emissions from Freight Transportation	\$147 M	\$95 M
Avoided CAC Emissions from Freight Transportation	\$14 M	\$7 M
Total	\$161 M	\$102 M

Economic Competitiveness Benefits

Avoided External Congestion Costs

METHODOLOGY

External highway use costs refer to the monetized impact of congestion resulting from highway travel. By avoiding diverting some rail traffic to truck VMT, the Project is expected to reduce total network VMT and the social costs of road use.

These benefits were estimated using unit values of external highway use (for congestion, noise, and safety) provided in Table A-14 of the U.S. DOT BCA Guidance. These values represent the congestion cost per VMT for trucks in rural areas. To estimate the benefit of the reduced social cost of road use, these values are multiplied by the total VMT diverted to trucks in the No Build scenario.

⁵ <https://www.aar.org/article/freight-rail-moving-miles-ahead-on-sustainability#:~:text=Thanks%20in%20part%20to%20these,to%20move%20freight%20over%20land.>

⁶ <https://nacfe.org/research/affs/>

ASSUMPTIONS

Table 15 presents the assumptions used in the estimation of avoided external congestion costs.

Table 15: Assumptions Used in the Estimation of Congestion Benefits

Variable	Units	Value	Source
External Congestion Cost, Buses and Trucks, Rural	2022\$/vehicle-mile	\$0.08	US Department of Transportation, Benefit-Cost Analysis Guidance for Discretionary Grant Programs, December 2023.

Avoided Fuel Costs

METHODOLOGY

Fuel use costs refer to the monetized impact of additional fuel usage resulting from additional highway travel incurred in the No Build scenario. By avoiding diverting some rail traffic to truck VMT and avoiding additional rail miles, the Project is expected to reduce total gallons of fuel consumed related to freight transportation.

These benefits were estimated by calculating the fuel consumed per year based on 11.5 miles per gallon (mpg) fuel efficiency for diesel freight trucks and 500 ton-miles per gallon efficiency for freight train fuel efficiency. These were applied to the truck miles driven and ton-miles railed to calculate annual fuel consumption. Annual fuel consumption was then multiplied by the average cost of diesel per gallon. The price per gallon was calculated using the Energy Information Administration's Annual Energy Outlook 2023 fuel forecasts to escalate out to 2050.

ASSUMPTIONS

The fuel efficiency assumptions used in the analysis are presented in **Table 16**, while fuel costs are presented in **Section 0**.

Table 16: Assumptions Used in the Estimation of Fuel Cost Savings

Variable	Units	Value	Source
Average Heavy-Duty Diesel Vehicle Fuel Economy	miles/gallon	11.5	Maximum found in Annual Fleet Fuel Study conducted by the North American Council for Freight Efficiency
Train Fuel Efficiency	ton-miles/gallon	500	Average reported by the Association of American Railroads

Avoided Freight Transportation Costs

METHODOLOGY

The Project provides users of the Ports of Yellow Bend and Tensas with additional modes to transport goods to and from the port. As the Project connects to a wider network of rail, avoided transportation costs are only calculated based on how far the freight would have traveled to a transfer point if the rail connection was not operational.

ASSUMPTIONS

The Demand Projections section contained the tables showing expected commodities to be shipped, their volumes, and distance traveled. It was conservatively assumed that only half the freight volumes would shift to rail and the distance traveled was also reduced by the same amount. Assumptions used to calculate shipping costs are shown in **Table 17**.

Table 17: Assumptions for Cost of Shipping by Truck and Rail

Variable	Units	Value	Source
Average cost of shipping by Truck	2022\$/ton-mile	\$5.21	BLS Table 3-21: Average Freight Revenue Per Ton-mile (Current cents), accessed 2022 dollars.
Average cost of shipping by Rail	2022\$/ton-mile	\$0.019	BLS Table 3-21: Average Freight Revenue Per Ton-mile (Current cents), accessed 2022 dollars.

Estimate of Economic Competitiveness Benefits

Table 18 presents the estimates of the economic competitiveness benefits. The estimated discounted present value over the projected 30-year benefit period is \$1,498 million. As the avoided freight transportation costs are a significant part of the overall benefits, sensitivity analysis will include a complete removal of the monetary benefits of reduced costs of shipping.

Table 18: Estimate of Benefits of Fuel Cost Savings

Values in 2022\$	Undiscounted	Discounted
Avoided Fuel Costs	\$37 M	\$18 M
Avoided Roadway Congestion Costs	\$24 M	\$12 M
Avoided Additional Freight Transportation Costs	\$2,935 M	\$1,466 M
Total	\$2,996 M	\$1,496 M

State of Good Repair Benefits**Avoided Pavement Damage Costs****METHODOLOGY**

In estimating the pavement maintenance costs, the total mileage traveled by trucks in transporting freight in the No Build and Build cases are multiplied by the pavement maintenance cost factor.

ASSUMPTIONS

Table 19 presents the assumptions used to estimate the avoided pavement maintenance costs.

Table 19: Assumptions Used in the Estimation of Avoided Pavement Maintenance Costs

Variable	Units	Value	Source
Highway Pavement Damage Cost	2022\$/vehicle-mile	\$0.09	Assuming 60 kip 5-axle Comb/Rural Interstate. Data based on Addendum to the 1997 Federal Highway Cost Allocation Study Final Report, May 2000. Inflated to 2022\$

Maintenance Costs

METHODOLOGY

The Project is adding new track which will increase operations and maintenance costs. These will be treated as a disbenefit. As such the increased maintenance costs are based on the difference in maintenance and inspection costs between the No Build and Build scenarios.

ASSUMPTIONS

Table 20 highlights the assumptions used in the estimation of the maintenance costs. The table of cost estimates was developed by the project team and then converted to 2022 dollars.

Table 20: Assumptions used in the Estimation of Maintenance Costs in 2022 \$

Year	Vegetation	Signal	Surfacing	Ballast	Ties	Annual Total
1	\$29,100	\$14,550				\$43,650
2	\$29,100	\$14,550				\$43,650
3	\$29,100	\$14,550				\$43,650
4	\$29,100	\$14,550	\$24,250			\$67,900
5	\$29,100	\$14,550	\$24,250			\$67,900
6	\$29,100	\$14,550	\$256,080	\$388,000		\$687,730
7	\$29,100	\$14,550	\$24,250			\$67,900
8	\$29,100	\$14,550	\$24,250			\$67,900
9	\$29,100	\$14,550	\$256,080	\$388,000		\$687,730
10	\$29,100	\$14,550	\$24,250			\$67,900
11	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
12	\$29,100	\$14,550	\$256,080	\$388,000	\$630,351	\$1,318,081
13	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
14	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
15	\$29,100	\$14,550	\$256,080	\$388,000	\$630,351	\$1,318,081
16	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
17	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
18	\$29,100	\$14,550	\$256,080	\$388,000	\$630,351	\$1,318,081
19	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
20	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
21	\$29,100	\$14,550	\$256,080	\$388,000	\$630,351	\$1,318,081
22	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
23	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
24	\$29,100	\$14,550	\$256,080	\$388,000	\$630,351	\$1,318,081
25	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
26	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
27	\$29,100	\$14,550	\$256,080	\$388,000	\$630,351	\$1,318,081
28	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
29	\$29,100	\$14,550	\$24,250		\$630,351	\$698,251
30	\$29,100	\$14,550	\$256,080	\$388,000	\$630,351	\$1,318,081

Year	Vegetation	Signal	Surfacing	Ballast	Ties	Annual Total
Total	\$873,000	\$14,550	\$2,741,220	\$3,492,000	\$12,607,015	\$20,149,735

Estimate of State of Good Repair Benefits

Table 21 shows the estimates of the state of good repair benefits. The estimated discounted present value over the projected 30-year benefit period is \$1 million.

Table 21: State of Good Repair Benefit Estimates

Values in 2022\$	Undiscounted	Discounted
Avoided Pavement Damage Costs	\$28 M	\$14 M
Increased Maintenance Costs	-\$27 M	-\$13 M
Total	\$1 m	\$1 M

Summary of Findings and Benefit-Cost Outcomes

Table 22 and **Table 23** summarize the BCA findings. Annual costs and benefits are computed over the lifecycle of the Project (24 years). As previously stated, construction is expected to be completed by 2028 with some minor close-out activities occurring in early 2029 that would not have any transportation impacts. Benefits accrue during the full operation of the Project, starting in 2029.

Table 22: Benefit Estimates by Merit Criteria

Merit Criteria	Benefit Category	Undiscounted	Discounted
Safety	Avoided Freight Transportation Accidents	\$242 M	\$121 M
Environmental	Avoided GHG Emissions	\$147 M	\$95 M
Sustainability	Avoided CAC Emissions	\$14 M	\$7 M
Economic	Avoided Fuel Costs	\$37 M	\$18 M
Competitiveness	Avoided Roadway Congestion Costs	\$24 M	\$12 M
	Avoided Additional Freight Transportation Costs	\$2,935 M	\$1,466 M
State of Good Repair	Avoided Pavement Damage Costs	\$30 M	\$14 M
	Change in Maintenance Costs	-\$27 M	-\$13 M
Total		\$3,400 M	\$1,720 M

Table 23: Overall Results of the Benefit Cost Analysis, Millions of 2022 Dollars

Evaluation Metrics	Undiscounted	Discounted
Total Benefits	\$3,400 M	\$1,720 M
Total Costs	\$172 M	\$144 M
Net Present Value (NPV)	\$3,228 M	\$1,576 M
Return on Investment (ROI)	1877%	1096%
Benefit-Cost Ratio (BCR)	19.8	12.0
Payback Period (years)	9.0	10.0
Internal Rate of Return (IRR)	28.9%	

Considering all monetized benefits and costs, the estimated internal rate of return of the Project is 28.9 percent. With a 2.0 percent real discount rate for CO₂-related impacts and 3.1 percent real discount rate for all other impacts, the \$144 million investment would result in \$1,720 million in total benefits for a NPV of \$1,576 million and a BCR of approximately 12.0.

Benefit Cost Analysis Sensitivity

The BCA outcomes presented in the previous sections rely on many assumptions and long-term projections, all of which are subject to considerable uncertainty.

The primary purpose of the sensitivity analysis is to help identify the variables and model parameters whose variations have the greatest impact on the BCA outcomes: the “critical variables.”

The sensitivity analysis can also be used to:

- Evaluate the impact of changes in individual critical variables—how much the final results would vary with reasonable departures from the “preferred” or most likely value for the variable; and
- Assess the robustness of the BCA and evaluate, in particular, whether the conclusions reached under the “preferred” set of input values are significantly altered by reasonable departures from those values.

The outcomes of the quantitative analysis for the Project, using a 2.0 percent discount rate for CO₂-related impacts and 3.1 percent discount rate for all other impacts, are summarized in **Table 24**.

As the benefits are driven by the impacts associated with expected outage, the sensitivity analysis considers the impacts from the outage durations. In addition, a sensitivity analysis for the capital cost, and varying volumes was conducted as well.

Table 24: Sensitivity Analysis Results

Original NPV (Discounted)	Original BCR	Parameters	Change in Parameters	New NPV (Discounted)	Change in NPV	New BCR
\$1,576 M	12.0	Change in Capital Costs	Increase costs by 25%	\$1,541 M	-2.2%	9.6
			Decrease costs by 25%	\$1,612 M	2.2%	15.9
		No shipper cost efficiencies / freight transport savings	Shipper efficiency toggle set to 0	\$110 M	-93%	1.8
		Change in % of freight diverted	Volume diverted to rail set at 25% for all commodities	\$710 M	-55%	5.9

Based on the sensitivity analysis, a 25 percent change in the capital costs is expected to result in a +/- 2.2 percent change in the NPV with a BCR ranging from 9.6 to 15.9. Removing the freight transport savings reduces the NPV to \$110 million with a BCR of 1.8.

Assuming only 25% of the freight volumes are diverted from truck to rail reduces the NPV by 55% to \$710 million with a BCR of 5.9.

Supplementary Tables

This section presents additional assumptions used within the BCA. A complete list of assumptions used in the analysis is also available within the accompanying BCA model.

Freight Emission Factors – Trucks

Emissions per Gallon of Fuel Burned - Trucks (grams/ton-miles)						Source/Comment
Year	NOx	VOC	PM	SO ₂	CO ₂	
2022	0.0	0.00	0.003	0.0002	55	Based on MOVES average annual emission factors for trucks in Alaska. Assuming trucks traveling 30 miles per hour. Moves model run in April 2024.
2023	0.0	0.00	0.002	0.0002	54	
2024	0.0	0.00	0.002	0.0002	53	Conversion to ton-miles based on 6.3 mpg and 134 gallons per ton-mile assumption.
2025	0.0	0.00	0.002	0.0002	52	
2026	0.0	0.00	0.002	0.0002	51	
2027	0.0	0.00	0.001	0.0002	50	
2028	0.0	0.00	0.001	0.0002	50	
2029	0.0	0.00	0.001	0.0002	49	
2030	0.0	0.00	0.001	0.0002	48	
2031	0.0	0.00	0.001	0.0002	48	
2032	0.0	0.00	0.001	0.0002	47	
2033	0.0	0.00	0.001	0.0002	47	
2034	0.0	0.00	0.001	0.0002	47	
2035	0.0	0.00	0.000	0.0002	46	
2036	0.0	0.00	0.000	0.0002	46	
2037	0.0	0.00	0.000	0.0002	46	
2038	0.0	0.00	0.000	0.0002	45	
2039	0.0	0.00	0.000	0.0001	45	
2040	0.0	0.00	0.000	0.0001	44	
2041	0.0	0.00	0.000	0.0001	44	
2042	0.0	0.00	0.000	0.0001	44	
2043	0.0	0.00	0.000	0.0001	44	
2044	0.0	0.00	0.000	0.0001	44	
2045	0.0	0.00	0.000	0.0001	44	
2046	0.0	0.00	0.000	0.0001	44	
2047	0.0	0.00	0.000	0.0001	44	
2048	0.0	0.00	0.000	0.0001	44	
2049	0.0	0.00	0.000	0.0001	43	
2050	0.0	0.00	0.000	0.0001	43	
2051	0.0	0.00	0.000	0.0001	43	
2052	0.0	0.00	0.000	0.0001	43	
2053	0.0	0.00	0.000	0.0001	43	
2054	0.0	0.00	0.000	0.0001	43	
2055	0.0	0.00	0.000	0.0001	43	
2056	0.0	0.00	0.000	0.0000	0	
2057	0.0	0.00	0.000	0.0000	0	
2058	0.0	0.00	0.000	0.0000	0	

Freight Emission Factors – Freight Rail

Emissions per Gallon of Fuel Burned - Freight Trains (grams/gallon)						
Year	NOx	VOC	PM	SO ₂	CO ₂	Source/Comment
2022	89.0	3.37	1.940	1.880	10,217	Data from United States Environmental Protection Agency, Office of Transportation and Air Quality, "Emission Factors for Locomotives", EPA-420-F-09-025, April 2009.
2023	84.0	3.16	1.843	1.880	10,217	
2024	79.0	2.95	1.649	1.880	10,217	
2025	74.0	2.74	1.552	1.880	10,217	
2026	69.0	2.63	1.455	1.880	10,217	
2027	65.0	2.42	1.358	1.880	10,217	
2028	61.0	2.21	1.261	1.880	10,217	
2029	57.0	2.11	1.067	1.880	10,217	
2030	53.0	2.00	0.970	1.880	10,217	
2031	49.0	1.79	0.970	1.880	10,217	
2032	46.0	1.68	0.873	1.880	10,217	
2033	43.0	1.58	0.776	1.880	10,217	
2034	40.0	1.47	0.679	1.880	10,217	
2035	37.0	1.37	0.679	1.880	10,217	
2036	35.0	1.26	0.582	1.880	10,217	
2037	33.0	1.26	0.582	1.880	10,217	
2038	31.0	1.16	0.485	1.880	10,217	
2039	29.0	1.16	0.485	1.880	10,217	
2040	28.0	1.05	0.388	1.880	10,217	
2041	28.0	1.05	0.388	1.880	10,217	
2042	28.0	1.05	0.388	1.880	10,217	
2043	28.0	1.05	0.388	1.880	10,217	
2044	28.0	1.05	0.388	1.880	10,217	
2045+	28.0	1.05	0.388	1.880	10,217	

Fuel Prices – Diesel

Year	Diesel \$/gallon	Source/Comment
2024	\$3.74	Diesel fuel prices estimated using retail price per gallon, less applicable state and federal taxes, and escalated using the Energy Information Administration's Annual Energy Outlook for 2023. Prices were kept constant beyond 2050.
2025	\$3.46	
2026	\$3.34	
2027	\$3.21	
2028	\$3.11	
2029	\$3.13	
2030	\$3.14	
2031	\$3.16	
2032	\$3.18	
2033	\$3.20	
2034	\$3.21	
2035	\$3.25	
2036	\$3.25	
2037	\$3.28	
2038	\$3.29	
2039	\$3.30	
2040	\$3.31	
2041	\$3.34	
2042	\$3.34	
2043	\$3.36	
2044	\$3.36	
2045	\$3.37	
2046	\$3.45	
2047	\$3.46	
2048	\$3.47	
2049	\$3.48	
2050+	\$3.48	